

Does the Linear A lexicon contain the stems of Cretan toponyms? A geographic control

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Abstract

That certain Linear A sign-groups alternate endings on a single stem is established, and several of the pairs examined here appear in the field's recent state of the art (Davis 2025) as evidence of Minoan suffixation: SU-KI-RI-TA / SU-KI-RI-TE-I-JA, U-TA-I-SI / U-TA-I-SE, A-MI-DA-O / A-MI-DA-U. This paper claims none of those pairs and asks a different question of the same data: whether the stems on which they are built are specifically the stems of Cretan toponyms, or whether the resemblance is what one would expect between two scripts sharing a syllabary and a phonotactics. The answer is measurable and turns on a control that has not been applied before. Searching the corpus for sign-groups containing the stem of a Cretan toponym attested in the Linear B of Knossos yields eighteen matches over twenty-three stems, fifteen of them longer than the stem and twelve behaving as parties to a transaction (heading a document or carrying a quantity); against the shuffled-syllable null preserving lengths, $p < 0.002$, and the result survives Bonferroni correction for the five tests performed and the exclusion of two-syllable stems (twelve cases, $p = 0.006$). The decisive control is geographic: repeated with eighteen Mycenaean toponyms from Pylos and Messenia, equally Mycenaean but not Cretan, the matches fall from eighteen to four, and the Cretan result also exceeds a null of Mycenaean words drawn at random ($p = 0.027$). The Linear A lexicon therefore resembles the names of the places of Crete and not Mycenaean in general. The ambiguity no test resolves is declared: Cretan toponyms are largely Pre-Greek words inherited from Minoan itself, so the association is compatible with two readings, onomastic formation on a place name and the simple membership of the toponym in the same lexicon. The control does not distinguish between them and does not claim to.

Keywords: Linear A, Cretan toponyms, Linear B, Pre-Greek substrate, null models, geographic control.

1. What is already established, and what is asked here

Davis (2025), in his survey of recent linguistics-based advances in the study of Linear A, gathers the available evidence for Minoan affixation and publishes a series of pairs showing alternation of ending on a single stem. Among them, and this must be said at the outset, are the two cases the first version of this paper presented as its own:

SU-KI-RI-TA / SU-KI-RI-TE-I-JA U-TA-I-SI / U-TA-I-SE A-MI-DA-O / A-MI-DA-U
JA-DI-KI-TU / JA-DI-KI-TE-TE

He likewise publishes the evidence for the prefix I- (DA-MA-TE / I-DA-MA-TE; A-SA-SA-RA-ME / JA-SA-SA-RA-ME), for the prefix A- (PA-RA-NE / A-PA-RA-NE; SI-KI-RA / A-SI-KI-

RA) and for polysyllabic prefixes such as I-NA- (I-NA-JA-RE / I-NA-JA-PA-QA), all of which this author had located distributionally without knowledge of that publication.

The existence of those pairs and those affixes is therefore not what is contributed here. The question of this paper is another and is not answered in that literature: **what those stems are stems of.** That Minoan suffixes is established; that the suffixed words are built on Cretan place names specifically, and not on Mycenaean lexicon in general, is a different claim, one that requires a control and that such a control can decide.

2. Data and method

Toponyms: the twenty-three Cretan place names attested in the Knossos tablets in Linear B (pa-i-to, ku-do-ni-ja, a-mi-ni-so, ti-ri-to, se-to-i-ja, ru-ki-to, su-ki-ri-ta, da-wo, ra-to, e-ko-so, qa-mo, ku-ta-to, do-ti-ja, ra-su-to, tu-ri-so, u-ta-no, pu-na-so, a-pa-ta-wa, ri-jo-no, wa-to, ka-ta-ra-i, si-ra-ro, a-ka).

Linear A corpus: GORILA in the form of the LinearA Explorer (Hogan 2022), with sign verification in SigLA. Signs are transliterated with Linear B values, following the convention of the field and without implying any assumption of homophony (Steele and Meißner 2017).

Criterion: a sign-group counts if it begins with the two-syllable stem of a toponym, is longer than that stem, and is not the toponym itself. It is also recorded whether it heads a document or is followed by a quantity, the criterion for a “party” used in the companion paper on names formed on plant words.

Null: shuffling of corpus syllables preserving the syllable count of each group, five hundred permutations.

3. Result

Fifteen groups, twelve of them behaving as parties, against a mean of 5.6 and a maximum of 13 for the null: $p < 0.002$.

group	occ.	on	party	context
A-KA-RU	3	a-ka	yes	HT 2: heads the document, with bulk oil
A-PA-RA-NE	2	a-pa-ta-wa (Aptara)	yes	HT 96a
A-KA-NU-ZA-TI	1	a-ka	yes	KN Zc 7
A-MI-DA-U	1	a-mi-ni-so (Amnisos)	yes	ZA 10a
A-MI-TA	1	a-mi-ni-so	yes	ZA 10b, with

group	occ.	on	party	context
				quantity 4
A-MI-DA-O	1	a-mi-ni-so	no	PH 31b
A-PA-KI	1	a-pa-ta-wa	yes	KN Zb 40
A-PA-DU-PA	1	a-pa-ta-wa	no	PK Za 12
A-KA-TA	1	a-ka	no	KN 32a
KA-TA-RE	1	ka-ta-ra-i	yes	KH 41, with quantity 3
PA-I-KI	1	pa-i-to (Phaistos)	yes	HT 62+73, with quantity 1
SU-KI-RI-TE-I-JA	1	su-ki-ri-ta (Sybrita)	yes	HT Zb 158b, alone on a vessel
U-TA-I-SE	1	u-ta-no	yes	KH 7b, before CYP+D ½
U-TA-I-SI	1	u-ta-no	yes	KH 16, before CYP+D ½
U-TA-RO	1	u-ta-no	yes	HT 116a: heads the document

Two observations on particular cases, without claiming the pairs. **SU-KI-RI-TE-I-JA** adds to the stem of Sybrita the ending -i-ja, which is the feminine ethnic ending in Mycenaean (ku-do-ni-ja, do-ti-ja), and it appears alone on a clay vessel, the canonical position for a mark of ownership or provenance. **U-TA-I-SE** and **U-TA-I-SI** both occur at Khania and both before the same commodity in the same quantity, CYP+D ½, which places the alternation in an identical administrative context.

One circumstance affecting the evidential value of SU-KI-RI-TE-I-JA must also be declared: the inscription is on a storage vessel, class Zb, and the administrative status of that class is disputed (Schoep 2002 excludes them from her study of the administration; others invoke the parallel of the inscribed stirrup jars of Linear B). If Zb inscriptions are not administrative, the reading as a mark of ownership stands, but the case cannot be counted among those behaving as parties.

4. The control, which is what this paper contributes

If the association were due to phonotactics shared between Minoan and Mycenaean Greek, any list of Mycenaean toponyms would give the same result. Repeated with eighteen toponyms from Pylos and Messenia (pu-ro, me-ta-pa, pa-ki-ja-ne, ka-ra-do-ro, ro-u-so, ri-jo, e-re-e and others), equally Mycenaean but not Cretan:

list	stems	matches
Cretan toponyms	23	**18**
non-Cretan toponyms	18	4
Mycenaean words at random (mean of 300 samples)	23	7.1

Cretan toponyms give **four times as many matches** as non-Cretan ones, and the result also exceeds the null of Mycenaean words drawn at random ($p = 0.027$).

The Linear A lexicon does not resemble Mycenaean in general: it resembles the names of the places of Crete. That is the claim the control sustains, and it is the one the literature on affixation does not make because it does not ask the question.

5. Robustness, and a rule that does not apply here

Stem requirement. Excluding the two-syllable toponyms, where a-ka sits — the shortest stem and hence the most prone to chance matches — twelve matches remain with $p = 0.006$.

Multiple comparisons. Five tests were performed (four semantic fields plus the repetition with an extended list). The Bonferroni threshold at 5% is $p < 0.010$, which this result (0.001) and the plant field (0.005) exceed, and the other three fields do not.

Duhoux's rule, and why it is not the right criterion here. Davis recalls that Duhoux established that an affix is credible when the words showing it share four or more consecutive syllabograms. Applied to the fifteen cases of this paper, none passes: the best, SU-KI-RI-TE-I-JA against su-ki-ri-ta, shares three, and the rest two. Two things are worth noting, however. First, applied to the twelve pairs Davis himself publishes, two pass: A-SA-SA-RA-ME / JA-SA-SA-RA-ME and DA-KU-SE-NE / DA-KU-SE-NE-TI share four, while the other ten share three or fewer, as he himself notes. Second, and this is the pertinent point: **the two sets compare different objects.** Davis's pairs compare two Linear A forms with each other, which is a comparison internal to the corpus and for which the rule is designed; the cases of this paper compare a Linear A form with a Linear B toponym, where the length of the shared stem is bounded by the length of the toponym, which has two or three syllables. That our cases give two does not measure the weakness of the resemblance but the brevity of the toponyms.

The equivalent quality criterion for this comparison is the geographic control of the previous section, and that is what is offered.

6. The shared suffixes

Comparing the endings of these fifteen cases with those of the nine in the companion paper on plant words, three suffixes appear on both classes of stem: -ja (SU-KI-RI-TE-I-JA against KU-PA-JA and KI-KI-RA-JA), -ti (A-KA-NU-ZA-TI against KA-NA-NI-TI) and -pa (A-PA-DU-PA against KA-KU-PA). The last two have one case per class and sustain nothing.

The case of -ja rests on three converging observations: it had already been measured as productive by another route (seven root/derivative pairs against 3.3 for the null); it appears on both classes of stem, indicating that it belongs to neither but to the language; and it coincides with the feminine ethnic suffix of Mycenaean. Searched for outside onomastics, there are twenty-eight groups ending in -ja and seven have their stem attested separately, of which only one belongs to our classes: **-ja is not an onomastic suffix but a general suffix of the language**. Davis (2025) independently documents the alternation of endings in several of these same groups.

7. Discussion

What holds. There is an association between the Linear A lexicon and the place names of Crete that is explained neither by chance nor by general resemblance to Mycenaean, and that a geographic control separates from both alternatives.

What does not hold. Any individual case. Thirteen of the fifteen have a single attestation. What exceeds the null is the set, and the set is what is offered.

And the underlying ambiguity, which no test resolves and which the control does not touch. Cretan toponyms are largely Pre-Greek words inherited from Minoan itself. That a Minoan sign-group should contain the stem of a Cretan toponym admits two readings: that it is a name formed on that toponym, or that the toponym belongs to the same Minoan lexicon and shares its phonological material. The geographic control distinguishes “resembles the places of Crete” from “resembles Mycenaean”, but not those two from each other, precisely because both predict the same thing. What would decide is a document in which the sign-group appeared alongside the toponym written separately, and that document is not in the corpus.

What this adds to what is published. Davis asks whether Minoan uses affixes, and the answer is that it does. This paper asks what the stems being affixed are stems of, and measures that they are specifically Cretan. Two questions of the same data, and the second requires a control the first does not need.

What would refute this work

This work must state what observation would refute it, under the third requirement of the protocol its author proposes. Two would. First, that a count using the full list of toponyms of Roman or Byzantine Crete, that is, Cretan place names Minoan cannot have generated, should yield a number of matches comparable to that of the Linear B toponyms: this would show that the asymmetry measures Cretan phonotactics in general and not Minoan inheritance. And second, that a systematic treatment of Linear A onomastics, such as the one occupying chapter 13 of van Soesbergen, should account for the fifteen cases by a route requiring no toponym stems, in which case the geographic control would remain a fact but would cease to sustain this reading.

Data

Cretan toponyms: the Knossos tablets in Linear B. Linear A corpus: GORILA (Godart and Olivier 1976-1985) in the form of the LinearA Explorer (Hogan 2022), with verification in SigLA (Salgarella and Castellan 2020). Scripts: the kuro package (Acedo 2026d).

Declaration of assistance

Corpus processing, the permutation tests and the draft of this text were produced with the assistance of a language model (Claude, Anthropic) under the author's direction, who is responsible for all claims.

References

- Acedo, A. 2026a. Minoan names formed on plant words. Draft.
- Acedo, A. 2026d. kuro: null models, calibration and confounder tests for small epigraphic corpora. Software.
- Beekes, R. S. P. 2010. Etymological Dictionary of Greek. Leiden.
- Davis, B. 2025. Recent linguistics-based advances in the study of Linear A. In E. Salgarella and V. Petrakis (eds), *The Wor(l)ds of Linear A*, AURA Supplement 15, Athens, 43-56.
- Duhoux, Y. 1978. Une analyse linguistique du linéaire A. In Y. Duhoux (ed.), *Études minoennes I*, Leuven, 65-129.
- Godart, L. and J.-P. Olivier 1976-1985. *Recueil des inscriptions en linéaire A*, I-V. Paris.
- Hogan, R. 2022. Linear A Explorer. <https://lineara.xyz>
- Hogan, R. 2025. Creating a network graph from the Linear A tablets. In E. Salgarella and V. Petrakis (eds), *The Wor(l)ds of Linear A*, AURA Supplement 15, Athens, 77-92.
- McArthur, J. K. 1993. Place Names in the Knossos Tablets. *Minos Suppl.* 9, Salamanca.
- Salgarella, E. and S. Castellan 2020. SigLA: the Signs of Linear A. <https://sigla.phis.me>
- Schoep, I. 2002. *The Administration of Neopalatial Crete*. *Minos Suppl.* 17, Salamanca.
- Steele, P. and T. Meißner 2017. From Linear B to Linear A: the problem of the backward projection of sound values. In P. Steele (ed.), *Understanding Relations Between Scripts*, Oxford, 93-110.
- Sommerschield, T., Y. Assael, J. Pavlopoulos, V. Stefanak, A. Senior, C. Dyer, J. Bodel, J. Prag, I. Androutsopoulos and N. de Freitas 2023. Machine Learning for Ancient Languages: A Survey. *Computational Linguistics* 49, 1-44.
- Younger, J. G. 2024. Linear A texts and inscriptions in phonetic transcription. *Academia.edu*.